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|----------|--|-------------------------------|
| 3 | <p>(a) (numerically equal to) quantity of (thermal) energy/heat to change state/phase of unit mass
 at constant temperature
 <i>(allow 1/2 for definition restricted to fusion or vaporisation)</i></p> | <p>M1
 A1 [2]</p> |
| | <p>(b) (i) at 70 W, mass s⁻¹ = 0.26 g s⁻¹
 at 110 W, mass s⁻¹ = 0.38 g s⁻¹</p> | <p>A1
 A1 [2]</p> |

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2015	9702	41

- (ii) 1. $P + h = mL$ or substitution of one set of values
 $(110 - 70) = (0.38 - 0.26)L$
 $L = 330 \text{ J g}^{-1}$ C1
C1
A1 [3]
2. either $70 + h = 0.26 \times 330$
or $110 + h = 0.38 \times 330$ C1
 $h = 17/16/15 \text{ W}$ A1 [2]